MYODRUM

A muscle-controlled virtual instrument using EMG and IMU sensing

[Author full name 1] | [Author full name 2] | Project 26P02

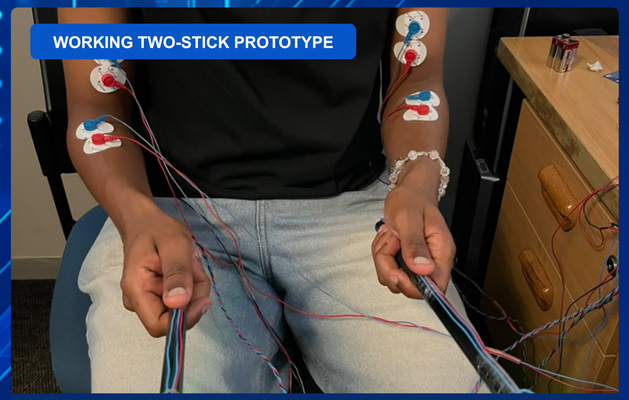
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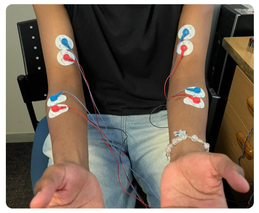
Why MyoDrum?

Can muscle activity and arm motion identify intentional air-drumming strikes and choose a virtual drum quickly enough for interactive play?

Goal: an affordable wearable instrument that turns natural drumming gestures into sound.



Two MPU6050 sensors + four active EMG channels



FOUR EMG CHANNELS

- A0 Left forearm
- A2 Left triceps
- A3 Right forearm
- A4 Right triceps

Six-channel expansion follows the four-channel comparison.

How MyoDrum works

EMG muscle activity → SYNCHRONIZED SENSOR WINDOW

IMU stick movement → SYNCHRONIZED SENSOR WINDOW

SYNCHRONIZED SENSOR WINDOW → CHOOSE A CLASSIFIER → PLAY SOUND

The IMU describes what the stick did. EMG helps decide whether the movement was intentional.

KICK • SNARE • CRASH

Rule-based classifier - working now

1. **IMU proposes**
   A fast movement, reversal and acceleration change create a strike candidate.
2. **EMG confirms**
   The active arm must cross its own calibrated muscle threshold.
3. **Safeguard**
   Cooldown and return travel suppress double or triple sounds.

Bagged-tree classifier - next stage

One labelled sensor window → Motion + muscle features → Many decision trees

Each tree votes: NO HIT | KICK | SNARE | CRASH

MAJORITY VOTE = FINAL PREDICTION

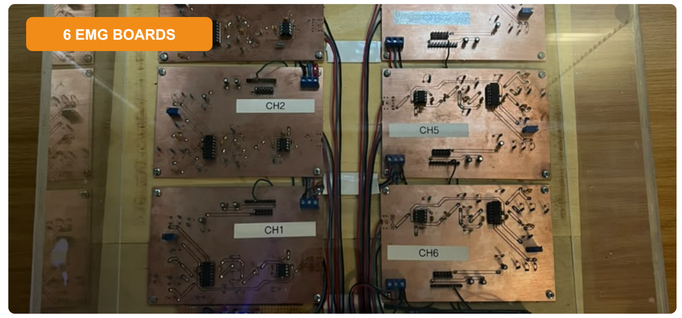
A group of trees should be less sensitive to one noisy electrode or one unusual swing than a single tree.

Virtual drum zones

SNARE left pose ← KICK centre / down → CRASH raised / right

Zones are defined relative to a calibrated neutral pose, not by double-integrated position.

Built hardware

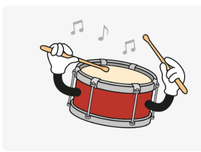


Signal conditioning, power and Arduino Mega 2560

Machine-learning data status

3	72	1,080
sessions	labelled blocks	requested hits

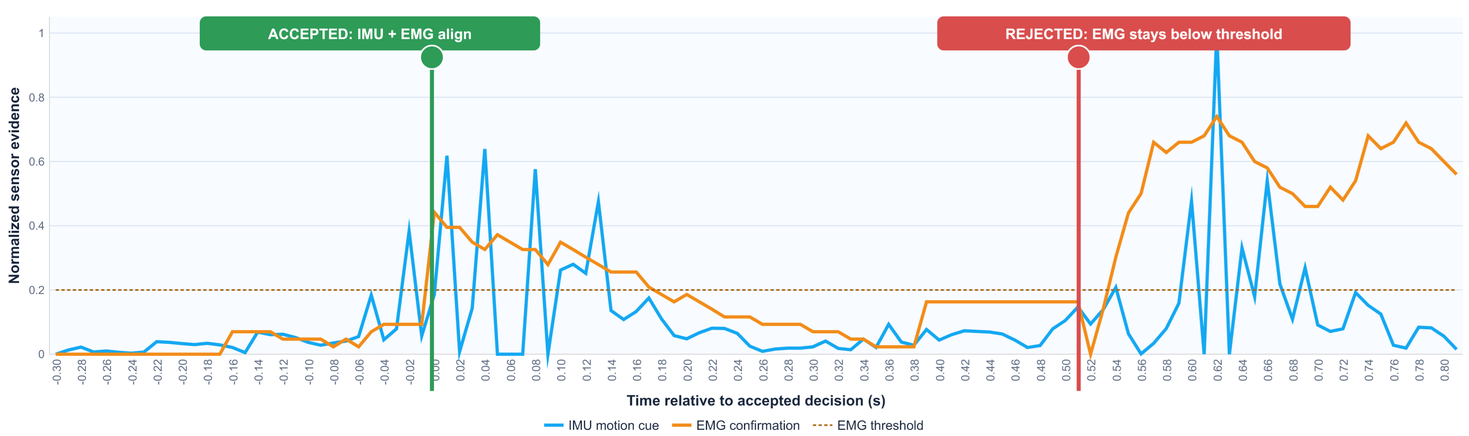
NEXT: TRAIN → TEST ON AN UNSEEN SESSION



WHAT STILL NEEDS TESTING

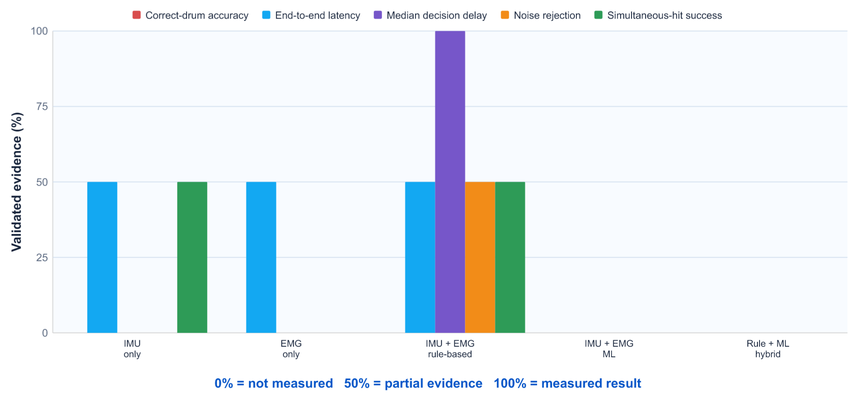
- held-out accuracy
- false-trigger rate
- sound latency
- orientation drift
- 6-channel benefit

Accepted versus rejected strike: synchronized IMU and EMG



Real right-hand events from session_20261012_165912. The accepted decision occurs at 0.00 s; the EMG-timeout rejection follows 0.51 s later.

Performance metrics by operating mode



0% = not measured 50% = partial evidence 100% = measured result

Current result: rule-based median decision delay = 0 ms (95th percentile = 125 ms).

Accuracy = correct drum selections ÷ intended strikes × 100%. A 0% evidence bar means unmeasured, not zero system accuracy.

CONCLUSION MyoDrum already combines synchronized motion and muscle sensing for three virtual drums. The next step is to train the bagged-tree model and compare it fairly with IMU-only, EMG-only and rule-based fusion.

References: TDK InvenSense, MPU-6050 Product Specification; SENIAM, Sensor Locations; MyoDrum recorded sessions. *Classifier delay excludes speaker/Bluetooth audio latency.